



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

USB TO ETHERCAT ADAPTER BOARD

The domain of the Project: PCB Designing

**Team Mentors (and their designation): Mr.Paramesh kumar
(Hardware Engineer)**

**By:
Ms.Singanavena Sahasra (ECE)**

Period of the project

March 2026 to April 2026



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Declaration

The project titled “USB To ETHERCAT Adapter Board” has been mentored by Paramesh kumar, organised by SURE Trust, from April 2026 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms.Sahasra

Mentor’s Name:Mr.Paramesh Kumar

Designation—Company Name: Hardware Design Engineer at Wakefit Innovation

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

The USB to EtherCAT Adapter Board is a PCB-based communication interface designed to provide connectivity between a USB host system and an EtherCAT industrial communication network. The main objective of the project is to develop a compact and reliable hardware platform that can be used to connect a computer or embedded system to EtherCAT-based industrial devices.

The project involves understanding USB and EtherCAT communication interfaces, selecting suitable electronic components, developing the schematic, assigning appropriate footprints, designing the PCB layout, and verifying the design according to PCB design practices.

The board is intended to provide USB connectivity on one side and an EtherCAT network interface on the other side. The overall architecture consists of the USB interface, processing/interface circuitry, EtherCAT communication circuitry, Ethernet physical-layer interface, power-supply section, protection circuitry, and required connectors. The project provides practical experience in PCB schematic design, component selection, footprint assignment, PCB routing, signal integrity considerations, power distribution, design-rule checking, and hardware interfacing. The completed design can serve as a foundation for USB-based EtherCAT communication in industrial automation, testing, development, and laboratory applications.



Introduction

Background

Industrial automation systems require reliable and high-speed communication between controllers, sensors, actuators, drives, and other field devices. EtherCAT is an industrial Ethernet-based communication technology used for real-time communication in automation systems. Computers and development systems commonly provide USB interfaces, while many industrial automation devices communicate through EtherCAT. Therefore, an interface is required to bridge these two communication environments.

The USB to EtherCAT Adapter Board is proposed as a hardware solution for this requirement. The PCB provides the necessary electronic interfaces and supporting circuitry to connect a USB host system to an EtherCAT network.

Problem Statement

Many computers and general-purpose embedded systems do not directly provide an EtherCAT interface. This creates a requirement for an external communication adapter.

The problem addressed by this project is the design of a compact PCB that can interface a USB host with an EtherCAT network while providing appropriate power management, signal routing, connectors, and protection circuitry.

Project Scope

- Understanding USB communication requirements.
- Understanding EtherCAT communication requirements.
- Designing the complete circuit schematic.
- Selecting suitable electronic components.
- Assigning component footprints.
- Designing the PCB layout.
- Routing high-speed communication signals.
- Designing the power-supply section.
- Providing suitable connectors.
- Adding protection circuitry wherever required.
- Performing Electrical Rules Check (ERC).
- Performing Design Rules Check (DRC).
- Generating manufacturing outputs such as Gerber files.
- Preparing the final PCB for fabrication and testing.



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Limitations

The actual communication performance depends on the selected USB interface/controller, EtherCAT controller or interface IC, Ethernet PHY, PCB stack-up, routing quality, power supply, firmware/software support, and connected EtherCAT devices.

Hardware design alone does not guarantee complete EtherCAT functionality; appropriate firmware, drivers, and host-side software may also be required.

Innovation

The project combines a commonly available USB host interface with an industrial EtherCAT communication interface into a dedicated PCB.

The board can provide a compact hardware platform for development, testing, and integration of USB-based systems with EtherCAT networks.



Project Objectives

Main Objective

To design and develop a reliable USB to EtherCAT Adapter Board that enables communication between a USB host system and an EtherCAT industrial network.

Specific Objectives

1. To study USB and EtherCAT communication interfaces.
2. To understand the hardware requirements of an EtherCAT interface.
3. To select appropriate controller, PHY, power, protection, and connector components.
4. To develop the complete circuit schematic.
5. To assign suitable PCB footprints to all components.
6. To design the PCB layout according to good PCB design practices.
7. To properly route high-speed USB and Ethernet/EtherCAT-related signals.
8. To design appropriate power-supply and decoupling circuits.
9. To incorporate protection against electrical disturbances where required.
10. To perform schematic and PCB design verification using ERC and DRC.
11. To generate manufacturing files for PCB fabrication.
12. To test and validate the completed board.

4.3 Expected Deliverables

- Complete circuit schematic
- Component/BOM list
- PCB layout
- PCB 3D model
- Gerber manufacturing files
- Design-rule verification results
- Fabricated PCB
- Hardware testing results
- Project documentation
- GitHub/source files, if applicable



1. Methods / Technology Used:

- USB communication
- STM32F4 microcontroller
- LAN9252 EtherCAT controller
- I²C EEPROM
- Ethernet/EtherCAT interface
- Power supply and filtering
- PCB schematic and layout design
- ERC and DRC verification

2. Tools / Software Used:

- **KiCad** – schematic and PCB design
- **KiCad PCB Editor** – component placement and routing
- **KiCad 3D Viewer** – PCB visualization
- **ERC/DRC** – design verification
- **Component datasheets** – component selection and circuit design

3. Data Collection Approach:

Technical information was collected from component datasheets and reference specifications to determine component connections, voltage requirements, footprints, and circuit design requirements.

4. Project Architecture:

The architecture of the USB to EtherCAT Adapter Board consists of multiple functional blocks that work together to provide communication between a USB host and an EtherCAT network.

Architecture Description

USB Host:

A computer or other USB host is connected to the adapter board through the USB interface.

STM32F4:

The STM32F4 microcontroller serves as the main control and processing device. It interfaces with the supporting circuits and the EtherCAT communication section.

LAN9252:

The LAN9252 is the dedicated EtherCAT controller. It handles the EtherCAT communication functionality and connects the processing/control section to the network interface.



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EtherCAT Interface:

The EtherCAT network interface provides the physical connection to the external EtherCAT network through the network connectors.

PC EEPROM:

The EEPROM provides non-volatile memory for storing required configuration or identification information.

Power Supply:

The power section provides the required voltage rails, including the +5 V and +3.3 V rails shown in the schematic.

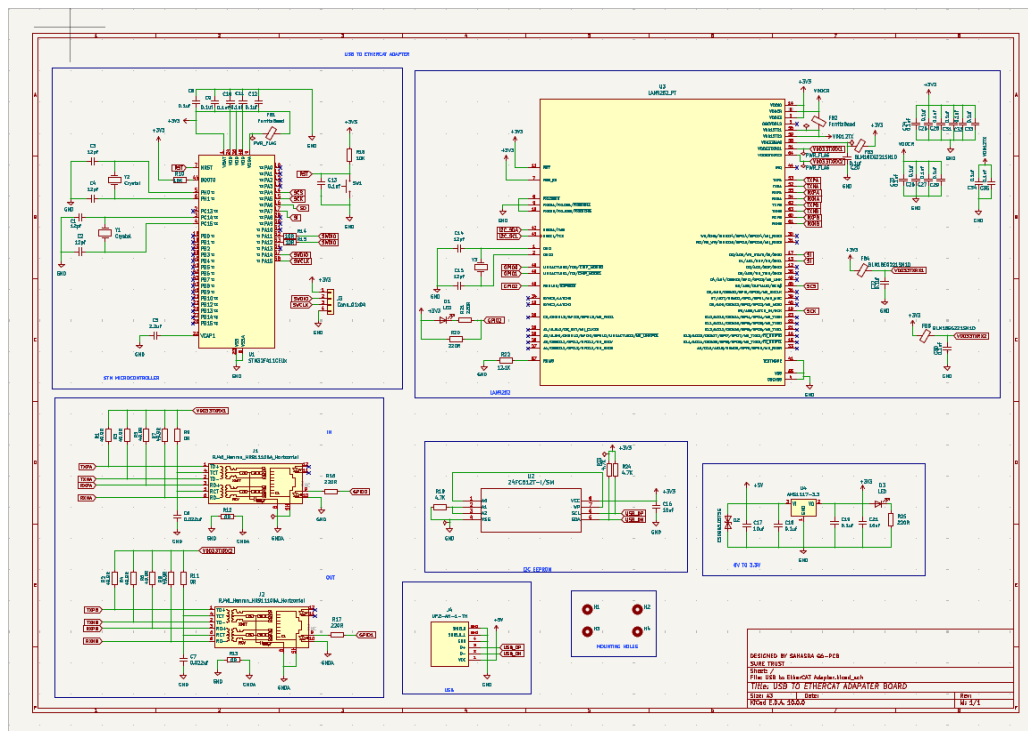
Clock Circuit:

Crystal oscillator circuits provide clock signals required by the controller devices.

Supporting Components:

Resistors, capacitors, filtering components, protection components, and connectors support the operation and reliability of the main functional blocks.

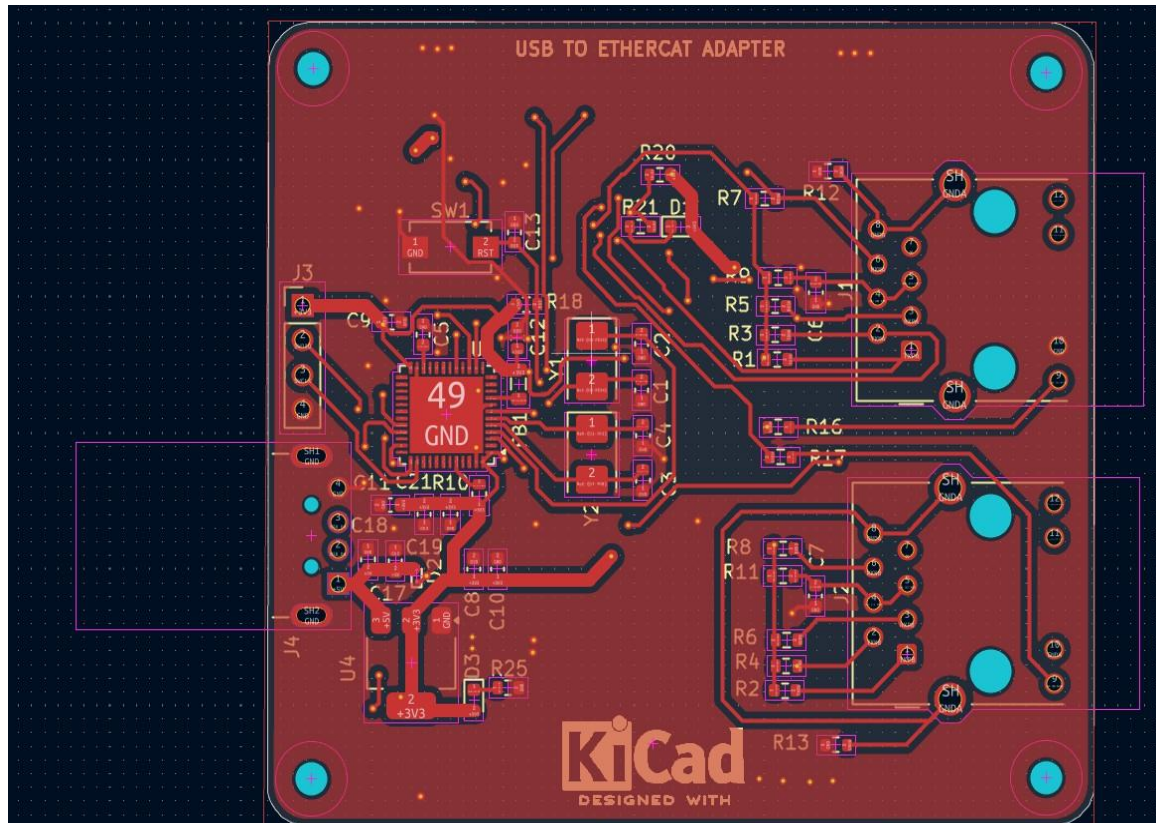
5.Final Project Screenshot and Supporting Explanation:



This schematic represents a hardware blueprint for a USB to EtherCAT Adapter Board, designed to enable a computer or host system to interface with industrial EtherCAT networks via USB. The core processing is handled by an STM32 microcontroller, which manages the system logic and interfaces directly with a dedicated LAN9252 EtherCAT controller chip that handles protocol processing. The board features dual RJ45 Ethernet ports configured as IN and OUT channels, allowing it to integrate seamlessly into standard EtherCAT daisy-chain topologies. Power is supplied via the USB connector and stepped down from 5V to 3.3V using an AMS1117-3.3 linear regulator to drive the onboard digital ICs. Additionally, a 24FC512T EEPROM chip stores the EtherCAT Slave Information (ESI) and system configuration settings over an I2C bus.



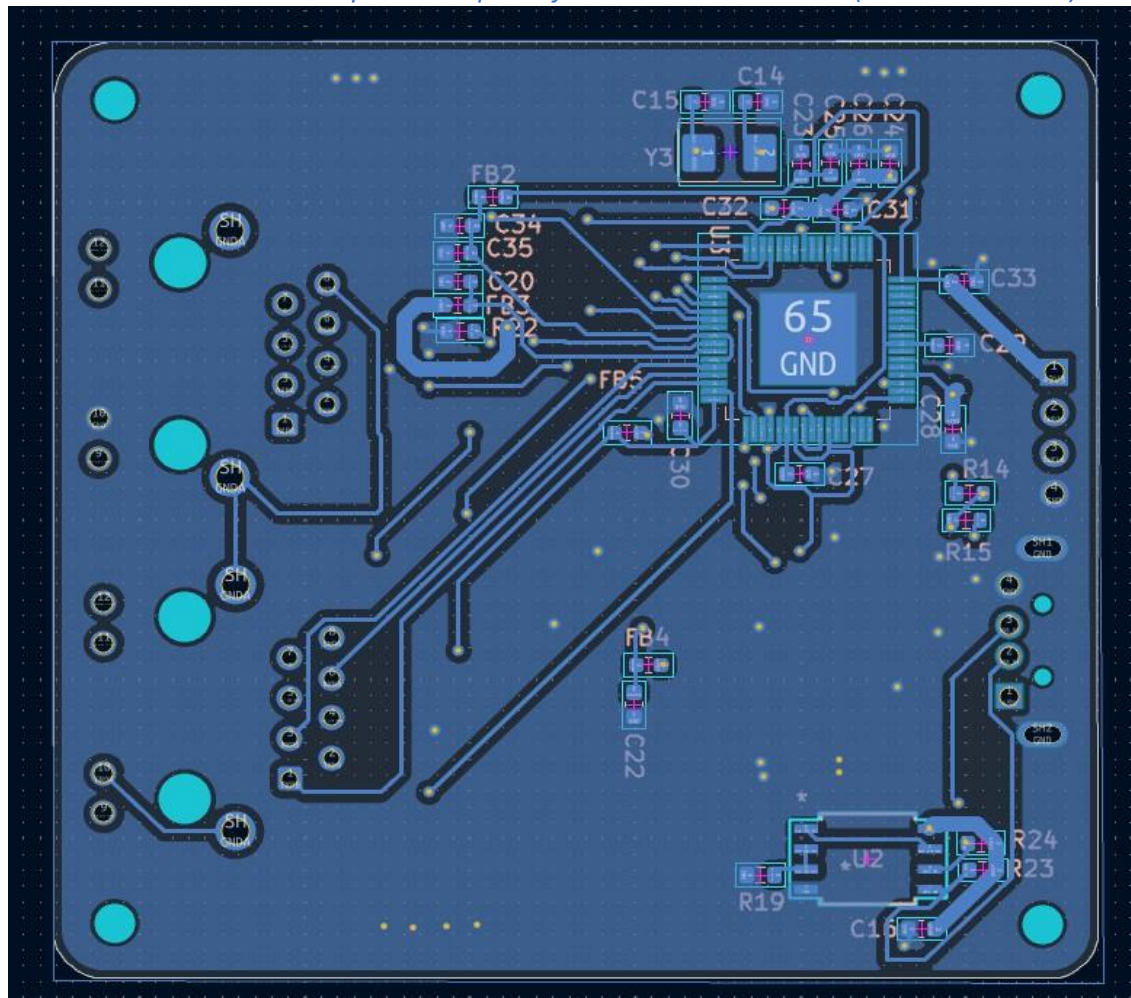
PCB Layout F.Cu(Front Copper Layer):



The front-layer PCB layout of the USB to EtherCAT Adapter Board was designed using KiCad. It shows the component placement, routed tracks, copper area, connectors, mounting holes, and silkscreen markings on the front side of the PCB. The layout was designed to provide proper connectivity and organized placement of the different circuit sections.

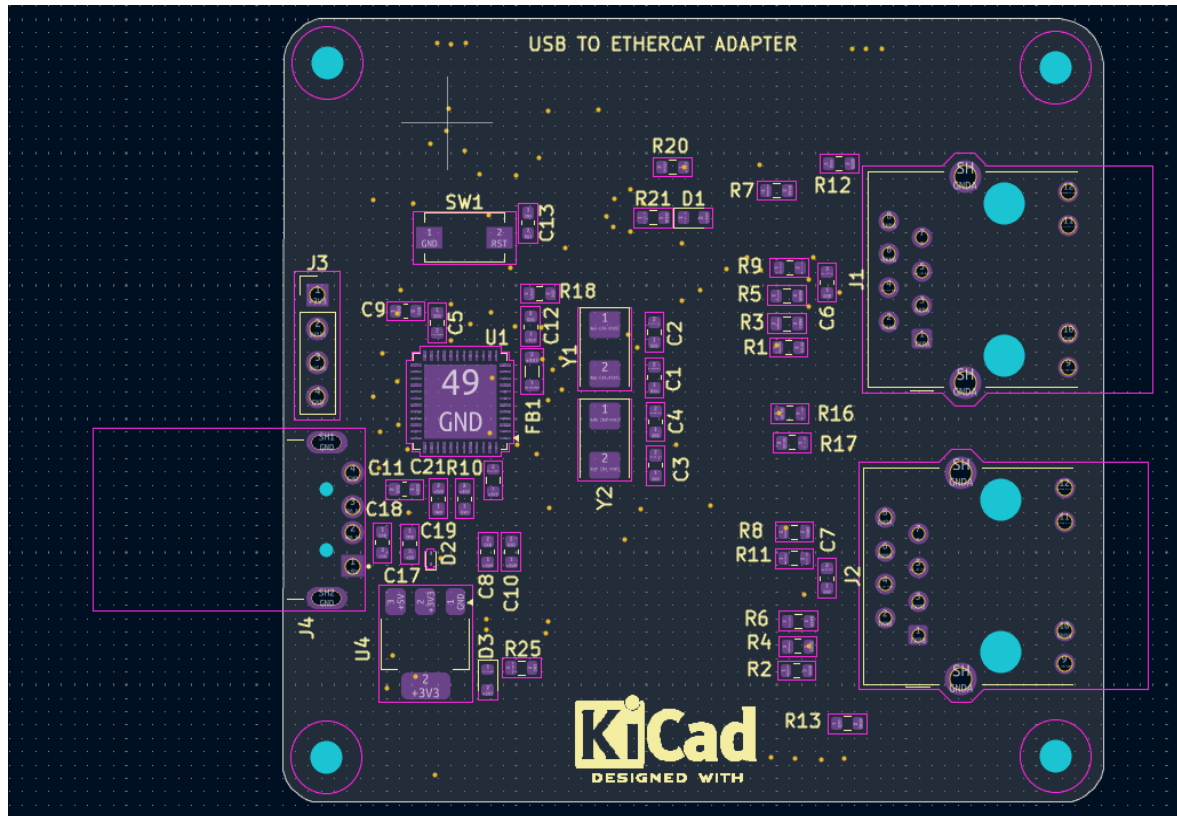
B.Cu (Back Copper Layer):

The back-layer PCB layout of the USB to EtherCAT Adapter Board shows the routed tracks, component connections, vias, mounting holes, and supporting circuitry on the back side of the PCB. The layout provides the required electrical connections between different circuit sections while maintaining an organized board design.



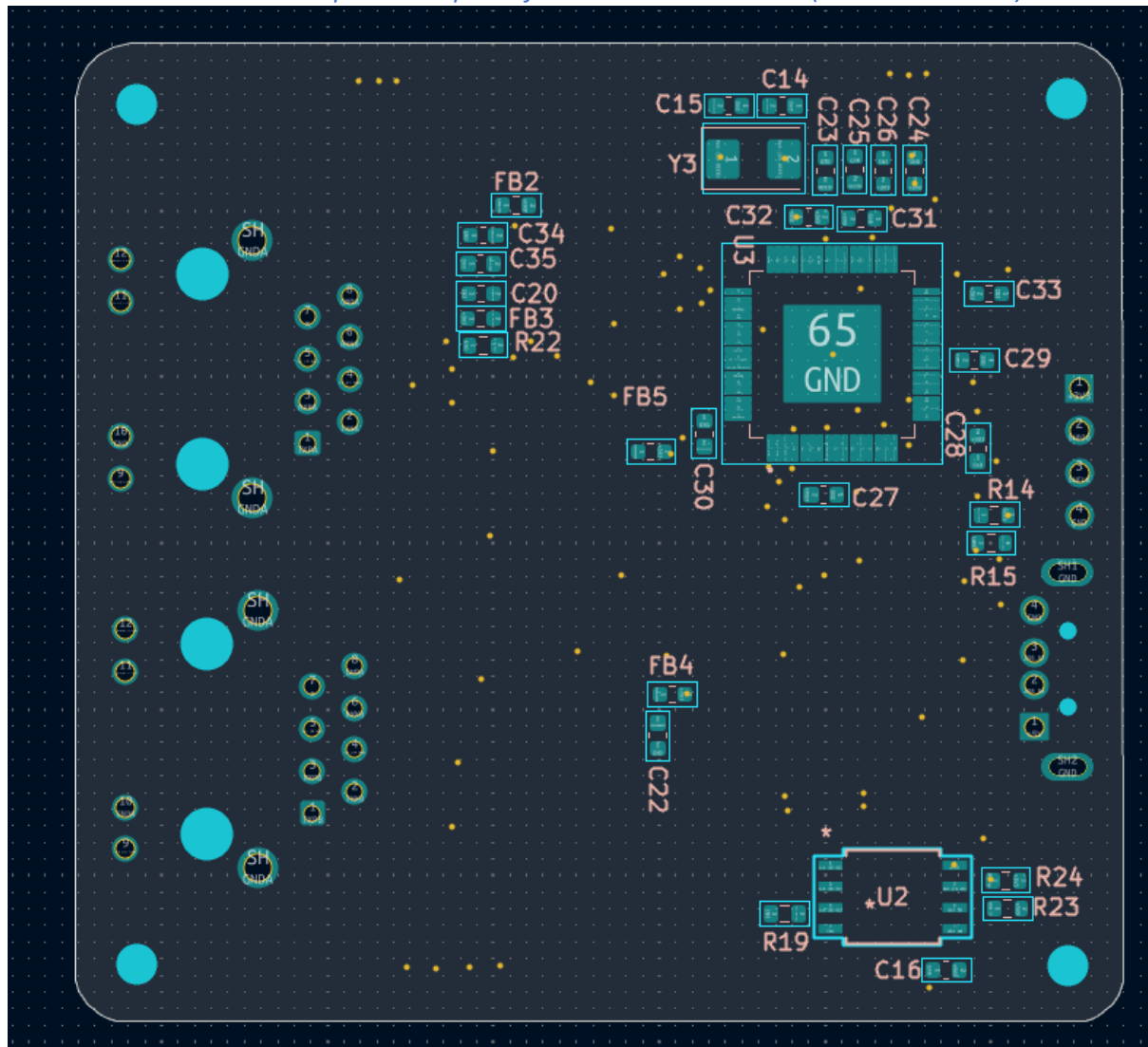
F.Assembly (Front Assembly Layer):

The front assembly view shows the placement and arrangement of components on the front side of the USB to EtherCAT Adapter Board. It includes the main IC, resistors, capacitors, connectors, switch, mounting holes, and other components along with their reference designators. This view helps verify the component placement and assists during PCB assembly and inspection.



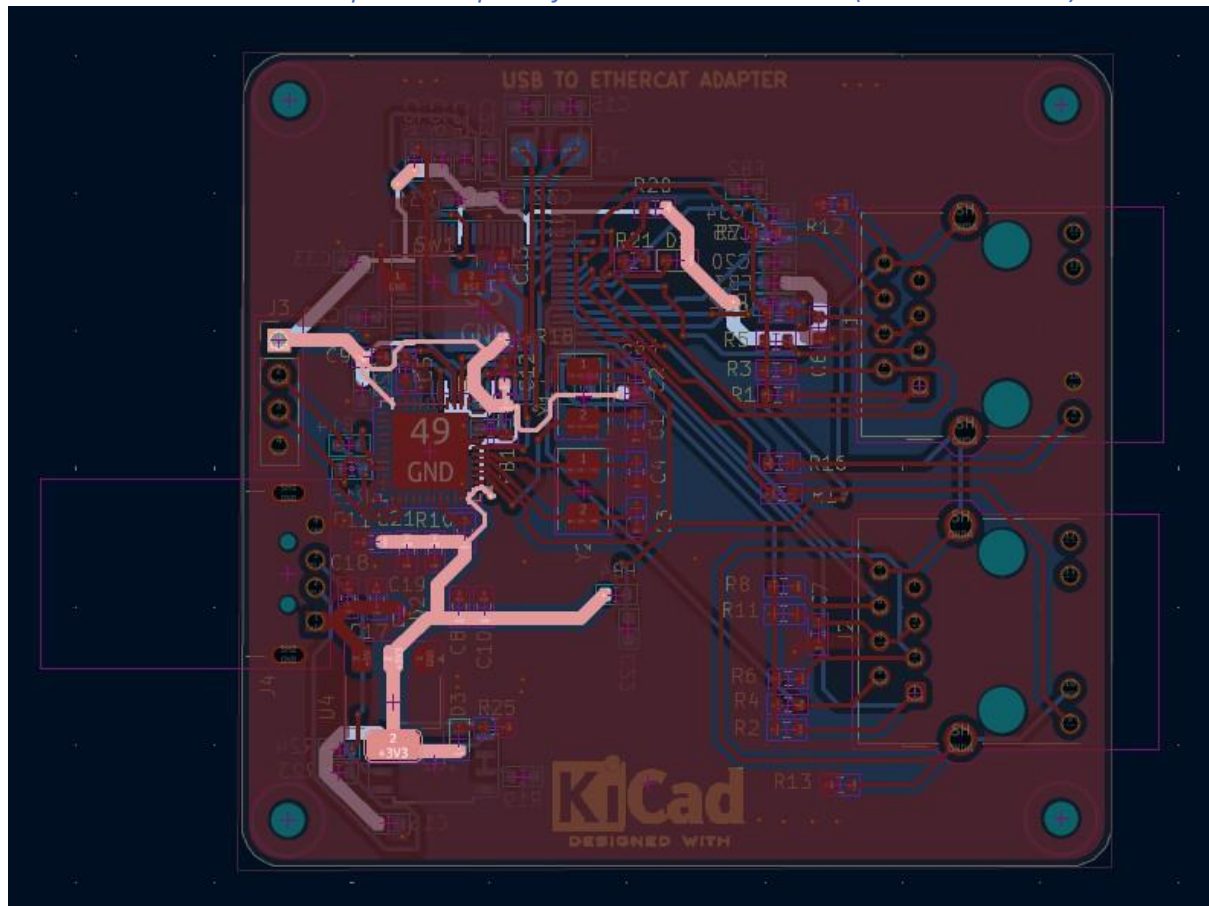
B.Assembly (Back Assembly Layer):

The back assembly view shows the components placed on the back side of the USB to EtherCAT Adapter Board. It displays the IC, resistors, capacitors, ferrite beads, and other components along with their reference designators. This view is useful for verifying the back-side component placement and for PCB assembly and inspection.



Trace Routing:

The trace routing view shows the copper tracks connecting the different components of the USB to EtherCAT Adapter Board. It includes power and signal traces routed between the major circuit sections, with wider traces used for power connections and controlled routing used for communication signals. The routing is designed to provide proper electrical connectivity between the components.



ERC Verification:

Electrical Rules Checker

Violations (12) Ignored Tests (4)

- Warning: Label connected to only one pin
Global Label '50'
- Warning: Label connected to only one pin
Global Label '12C_SDA'
- Warning: Label connected to only one pin
Global Label '12C_SCL'
- Warning: Symbol 'LAN9252_PT' not found in symbol library 'LAN9252_PT'
Symbol U3 [LAN9252_PT]
- Warning: Pin of Unconnected and Unconnected are connected

Show: ☐ All ☒ Errors 0 ☒ Warnings 12 ☒ Exclusions Save...

Delete Marker Delete All Markers Run ERC Close

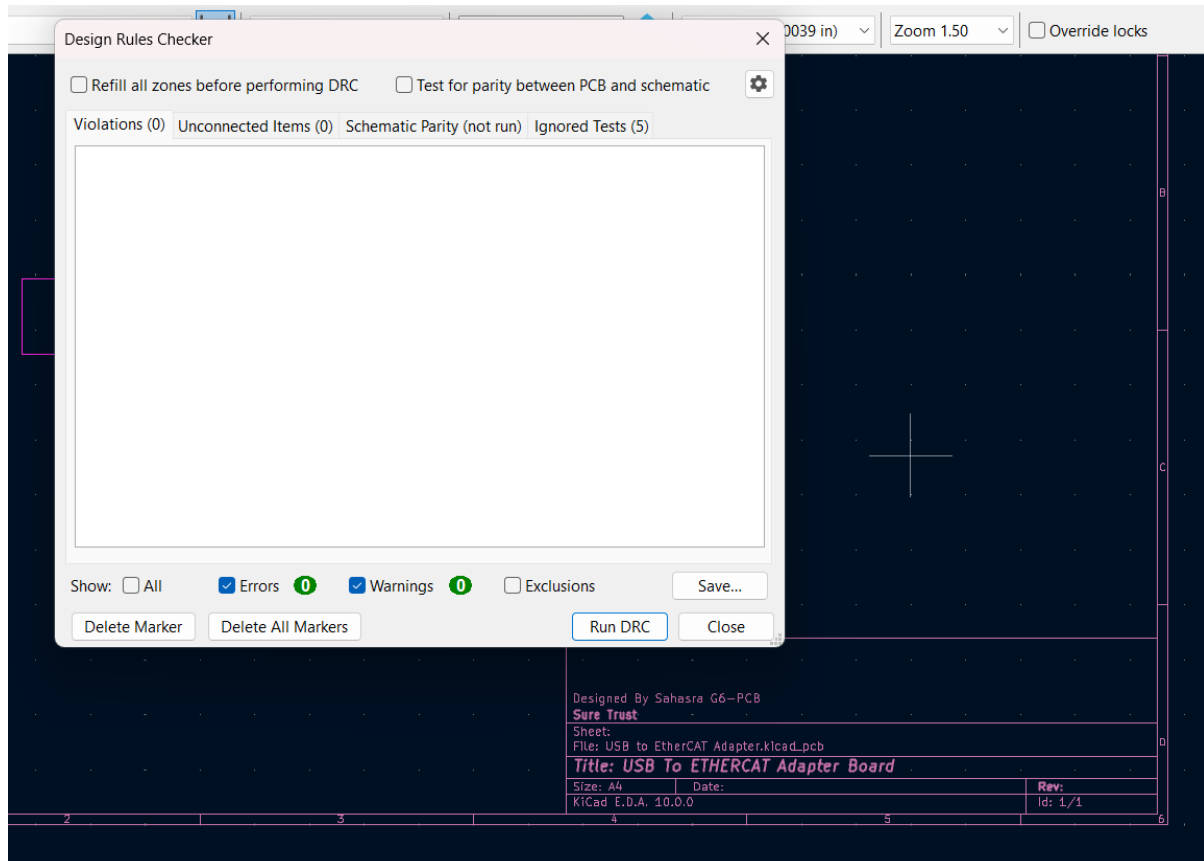
DESIGNED BY SAHASRA G6-PCB
SURE TRUST
Sheet: /
File: USB to EtherCAT Adapter.kicad_sch
Title: USB TO ETHERCAT ADAPTER BOARD
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.0 Id: 1/1



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The ERC check identifies 12 warnings but no errors, meaning the schematic does not have any major electrical rule violations. The warnings mainly indicate that labels such as SO, I2C_SDA, and I2C_SCL are connected to only one pin, so their connections should be verified. It also reports that the LAN9252_PT symbol is missing from the symbol library. These issues are mostly related to labeling and library configuration, and should be checked before the final PCB design.

DRC Verification :



Design Rule Check (DRC) was performed using KiCad to verify the PCB layout. The final design showed zero violations, zero unconnected items, zero errors, and zero warnings, indicating that the PCB layout satisfies the defined design rule



Learning and Reflection

Learning

During this project, we gained practical knowledge of:

- Designing schematics and PCB layouts using KiCad.
- Understanding the working of USB and EtherCAT communication.
- Working with the STM32F4 microcontroller and LAN9252 EtherCAT controller.
- Selecting components based on their datasheets and electrical requirements.
- Assigning suitable footprints to components.
- Performing component placement and trace routing.
- Understanding power and ground routing on a PCB.
- Using copper zones and vias effectively.
- Performing DRC verification and resolving design issues.
- Creating front and back assembly views.
- Understanding PCB manufacturing and assembly requirements.
- Adding logos and silkscreen markings to the PCB.
- Understanding the complete process from schematic → PCB layout → verification.

Overall Experience

This project provided us with valuable hands-on experience in PCB design and industrial communication hardware. Working on the USB to EtherCAT Adapter Board helped us understand how different circuit sections are integrated into a single PCB. We also improved our problem-solving, troubleshooting, teamwork, and time-management skills. The project gave us practical exposure to the challenges involved in PCB design and helped us gain confidence in working with professional PCB design tools.

Overall, the project was a useful and practical learning experience that strengthened our knowledge of PCB development and prepared us for future hardware and embedded-system projects.



Conclusion and Future Scope

Conclusion

The USB to EtherCAT Adapter Board project was successfully designed as a PCB-based interface for connecting a USB host system with an EtherCAT network. The project involved the design and integration of the STM32F4 microcontroller, LAN9252 EtherCAT controller, EEPROM, USB interface, power supply, and EtherCAT interfaces.

The project provided practical knowledge of schematic design, component selection, footprint assignment, PCB layout, trace routing, power routing, and PCB design verification using KiCad. The final PCB layout was completed with proper component placement, routing, mounting holes, and front and back assembly arrangements.

Overall, the project helped in understanding the complete process of developing a PCB, from circuit design to final board layout.

Future Scope

The project can be further developed in the following areas:

- PCB fabrication and assembly of the designed board.
- Testing the USB interface with a computer or host system.
- Testing EtherCAT communication with EtherCAT devices.
- Developing and integrating the required firmware for the STM32F4.
- Developing suitable PC/host-side software or drivers.
- Performing detailed hardware debugging and validation.
- Improving signal integrity for high-speed communication.
- Conducting EMI/EMC testing for industrial applications.
- Adding additional status LEDs and test points for easier debugging.
- Optimizing the PCB for compact size and mass production.
- Testing the board under different industrial operating conditions.
- Developing a suitable protective enclosure for practical deployment.